

Automated algorithm for mapping regions of cold-air pooling in complex terrain

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[1] In complex terrain, air in contact with the ground becomes cooled from radiative energy loss on a calm clear night and, being denser than the free atmosphere at the same elevation, sinks to valley bottoms. Cold-air pooling (CAP) occurs where this cooled air collects on the landscape. This article focuses on identifying locations on a landscape subject to considerably lower minimum temperatures than the regional average during conditions of clear skies and weak synoptic-scale winds, providing a simple automated method to map locations where cold air is likely to pool. Digital elevation models of regions of complex terrain were used to derive surfaces of local slope, curvature, and percentile elevation relative to surrounding terrain. Each pixel was classified as prone to CAP, not prone to CAP, or exhibiting no signal, based on the criterion that CAP occurs in regions with flat slopes in local depressions or valleys (negative curvature and low percentile). Along-valley changes in the topographic amplification factor (TAF) were then calculated to determine whether the cold air in the valley was likely to drain or pool. Results were checked against distributed temperature measurements in Loch Vale, Rocky Mountain National Park, Colorado; in the Eastern Pyrenees, France; and in Yosemite National Park, Sierra Nevada, California. Using CAP classification to interpolate temperatures across complex terrain resulted in improvements in root-mean-square errors compared to more basic interpolation techniques at most sites within the three areas examined, with average error reductions of up to 3°C at individual sites and about 1°C averaged over all sites in the study areas.

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1. Introduction

[2] *Whiteman et al.* [2001] define a cold-air pool as a “topographically-confined, stagnant layer of air that is colder than the air above.” The American Meteorological Society *Glossary of Meteorology* [*Glickman et al.*, 1999] (available at <http://amsglossary.allenpress.com/glossary>) adds, “This air can remain stagnant, trapped by the surrounding higher terrain, resulting in long periods of poor air quality and fog.” Cold pools can cause localized icing or freezing precipitation, and can delay the melting of snow and ice, thus having a large impact on basin hydrology. They are associated with regions of permafrost and provide unique microclimates that influence species distributions and diversity [*Tenow and Nilssen*, 1990; *Blennow and Lindkvist*, 2000; *Wearne and Morgan*, 2001]. Because of decoupling from the free atmosphere, these cold pool areas

may respond differently to climate change than surrounding regions. For the same reason, accurately representing the formation and persistence of cold-air pools in complex terrain is one of the most challenging forecast problems in many middle and high-latitude locales, including the United States [*Smith et al.*, 1997; *Jarvis and Stuart*, 2001; *Whiteman et al.*, 2001; *Stahl et al.*, 2006].

[3] Distributed snowmelt and ecological models require maps of distributed near-surface (i.e., approximately 2 m height) air temperature across a basin, and errors in interpolating temperature correctly across complex terrain are one of the largest sources of error in distributed modeling [*Singh*, 1991; *Chen et al.*, 1999; *Archer*, 2004; *Liston and Elder*, 2006]. These errors are particularly large in areas subject to cold-air pooling, where even high-resolution atmospheric models are often biased 3 to 4°C too warm during cold-pool events [*Hart et al.*, 2004].

[4] To address these issues, this article uses the results of several process-based studies to develop an automated algorithm for identifying regions of likely CAP that can be applied in any area of complex terrain where a digital elevation model (DEM) is available. Specifically, we use data from temperature sensors distributed about 2 m above the ground in the mountains of (1) the Rocky Mountains,

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Colorado, (2) the Pyrenees Orientales, France, and (3) the Sierra Nevada, California, to develop and check classification schemes for identifying areas of CAP based on DEMs. We then use one station in a known CAP region and one in a known no-CAP region to represent the temporal variance in CAP strength and then test if and how well CAP mapping improves temperature prediction above basic interpolation schemes in complex terrain. Developing models to predict the duration and strength of a CAP event is a separate subject for further research.

2. Previous Research

[5] Cold-air pools have been studied extensively with respect to temperature inversions and the dynamic meteorology of mountain valley winds [e.g., *Whiteman and McKee*, 1982; *Kondo et al.*, 1989; *Kondo and Okusa*, 1990; *Whiteman*, 1990; *O'Steen*, 2000; *Clements et al.*, 2003; *Clements and Zhong*, 2004; *Zangl*, 2005; *Steinacker et al.*, 2007]. Diurnal winds, forced by horizontal temperature gradients and their accompanying pressure gradients, are a regular part of mountain weather [*Whiteman*, 2000]. In general, winds flow upslope and up-valley during the day and downslope and down-valley at night. These winds, caused by different rates of heating and cooling of adjacent surfaces, are strongest when larger-scale winds are weak, and skies are clear [*Whiteman*, 2000].

[6] Slope winds are driven primarily by buoyancy forces. Following sunset, radiative energy loss cools air in contact with the surface. This cooler, denser air moves downslope and is replaced by warmer free air, which is cooled in turn. The volume of relatively warmer free air adjacent to a slope is what conserves the temperature of a slope. Therefore, exposed ridges and convexities, where there is a plentiful supply of free air, will not cool much, whereas concavities and areas cut off from the free atmosphere will cool more, with cold-air pooling occurring most frequently along flat valley bottoms in mountainous terrain [*Marvin*, 1914; *Barr and Orgill*, 1989; *Neff and King*, 1989; *Blennow*, 1998; *Gustavsson et al.*, 1998; *Whiteman et al.*, 1999; *Lindkvist et al.*, 2000; *Halley et al.*, 2003; *Chung et al.*, 2006].

[7] Valley winds flow along the longitudinal axis of a valley and are primarily driven by pressure gradients, which form as a result of along-valley temperature differences [*Whiteman*, 1990]. These along-valley temperature differences occur because different heating and cooling rates occur within different valley cross sections. Assuming the same net radiation flux occurs across the top of several valleys, a larger temperature change will occur within a valley with a smaller enclosed volume of air [*McKee and O'Neal*, 1989; *Whiteman*, 1990]. Thus, the diurnal temperature range in a valley is larger than that of an adjacent plain. This concept has been quantified by the topographic amplification factor, TAF, which for a valley cross section is defined as

$$\text{TAF} = \frac{\left[\frac{W}{A_{yz \text{ valley}}} \right]}{\left[\frac{W}{A_{yz \text{ plain}}} \right]} = \frac{\left[\frac{W}{A_{yz \text{ valley}}} \right]}{\left[\frac{1}{H} \right]}, \quad (1)$$

where W is the valley top width, A_{yz} is the cross-sectional area of the valley or equivalent plain, and H is the height of

the valley cross section, from its lowest point to the point where the top width is measured [*Wagner*, 1932, 1938; *Steinacker*, 1984; *McKee and O'Neal*, 1989; *Whiteman*, 1990; *De Wekker et al.*, 1998]. The TAF primarily affects differences in nighttime temperatures because daytime warming leads to unstable conditions and mixing. When the TAF decreases in the down-valley direction, air cools faster up-valley than down-valley, which leads to a horizontal pressure gradient that drives nocturnal down-valley winds. When the TAF increases in the down-valley direction, cooler air and higher pressure exist down-valley, which leads to stable conditions and cold-air pooling [*McKee and O'Neal*, 1989; *Whiteman*, 1990]. In addition to the TAF, valley constrictions (which can limit mass flux), cold air input from tributary valleys, and variations in sensible heat fluxes along valley walls also influence the diurnal temperature range observed in a valley [*Neff and King*, 1989; *De Wekker et al.*, 1998].

3. Methods

3.1. Observations for Validation: Distributed Temperature Data Sets

3.1.1. Study Locations and Instrumentation

[8] Temperature data sets from Loch Vale in the Colorado Rocky Mountains [*Lundquist and Rochford*, 2007] (available at <http://faculty.washington.edu/jdlund/home/publications.shtml>), from the Pyrenees, France [*Pepin and Kidd*, 2006], and from Yosemite National Park, Sierra Nevada, California [*Lundquist and Cayan*, 2007] were used to test the algorithm (Figure 1, Table 1). The Colorado study used Maxim 1922L iButtons [*Hubbart et al.*, 2005], and the Pyrenees and Sierra Nevada studies used Onset Tidbits and HOBOs [*Whiteman et al.*, 2000]. Instruments were deployed in evergreen trees approximately 2 m above the ground, a deployment method that compared well (root-mean-square error (RMSE) < 1°C) with nearby standard Gill-shielded temperature sensors on poles [*Lundquist and Huggett*, 2008]. These instruments have been successfully used in many studies [*Whiteman et al.*, 2001; *Taras et al.*, 2002; *Lookingbill and Urban*, 2003; *Lundquist et al.*, 2003; *Mahrt*, 2006; *Pepin and Kidd*, 2006; *Tang and Fang*, 2006; *Lundquist and Cayan*, 2007; *Marshall et al.*, 2007]. Table 1 details instrument specifications, sampling intervals, and accuracy, as well as topographic information, for each study area.

[9] The Loch Vale watershed [*Campbell et al.*, 2000; *Clow et al.*, 2003] in Rocky Mountain National Park, Colorado (Figure 1a), is a glaciated U-shaped valley. Site locations included flat valley bottoms, steeper stream-cut valleys, and the steep sidewalls of the valleys, all within the drainage area of the Loch (Figure 1a). A terminal moraine at the outlet of the Loch causes a terrain constriction which blocks cold-air drainage and leads to pooling above.

[10] The Eastern Pyrenees measurements [*Pepin and Kidd*, 2006] focused on transects across three river valleys draining a central plateau area (Figure 1b): the Conflent, which drains to the east/northeast and reaches the Mediterranean east of Perpignan, the Cerdagne, which drains southwest to Spain, and the Capcir, with flows north toward Carcassonne. The Cerdagne and Capcir are wide valleys

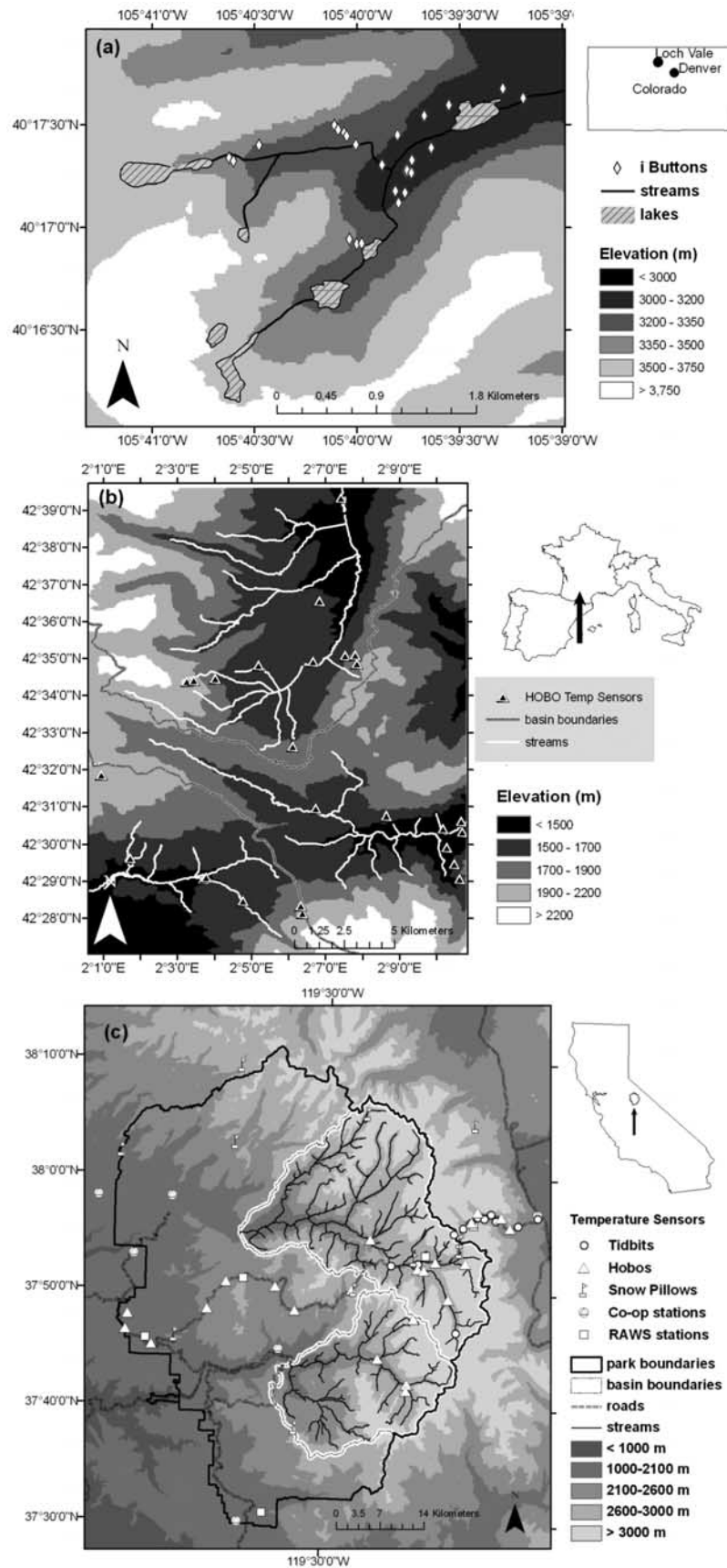


Figure 1. Maps of geographic locations of temperature sensors in (a) Loch Vale, Rocky Mountain National Park, Colorado, (b) Pyrenees, France, and (c) Yosemite, Sierra Nevada, California.

Table 1. Study Site Characteristics

	Rocky Mountains	Pyrenees	Sierra Nevada
Area of study (km ²)	15	400	7500
Elevation range (m)	3100–3400	1400–2200	1200–3200
Resolutions examined (m)	10, 20, 50, 100, 500	100, 500	100
Number of sites	17	26	51
Dates examined	Aug 2005 to Jul. 2006	May 2002 to May 2005	Jul 2002 to Jul 2005
Radius (half the average peak-to-peak distance) (m)	375	3500	1500
Instruments used	Dallas Semiconductor Maxim iButtons (DS-1922L, purchased 2005)	HOBO Pro Series RH and Temp (purchased 2002)	HOBO Pro Series RH and Temp (purchased 2003, 2004); onset StowAway TidbiT Temp Logger (purchased 2002, 2003, 2004)
Sampling interval (min)	60	15	30
Instrument response time (min)	<10	<10	<10
Instrument temperature range (°C)	–35–85	–30–50	–30–50 (HOBO), –20–50 (Tidbit)
Radiation shield design	upside-down white funnel, placed in tree	white PVC tubes, hung in evergreen trees at a 45° angle, with the top end facing north	rain shield from Onset Computer Corporation, painted brown, or Gill radiation shield; all shields placed in trees
Specified instrument accuracy (°C)	±0.5	±0.2	±0.3 (for HOBO Pro), ±0.5 (for Tidbit)
Deployed instrument/shield accuracy	better than ±1.0°C	better than ±1.0°C	better than ±1.0°C
Sensor microscale deployment/location	in evergreen trees, 2 m above the ground	in forested areas, away from paths and avoiding local topographic hollows	in evergreen trees, 2 m above the ground, generally along roads, trails, or streams

with flat bottoms, the latter being restricted in its lower reaches and particularly prone to cold air pooling. The Conflent is a V-shaped canyon with steep sides (20°–30°) and a steep longitudinal profile gradient.

[11] The Yosemite National Park, Sierra Nevada, California data set included not only HOBO loggers deployed in trees, but also RAWs sites, CA DWR snow pillow sites, and cooperative observing sites, as described by *Lundquist and Cayan* [2007]. Site locations spanned the eastern and western slopes of the central Sierra Nevada and ranged from glacier-carved U-shaped valleys to steep gorges to flatter meadows at the limits of tree line. Sensors were distributed along road corridors in addition to along streams, sampling undulating topography and escarpments in addition to fluvial- and glacial-carved valleys.

3.1.2. Empirical Orthogonal Function (EOF) Technique for Identifying Cold-Air Pooling (CAP) Locations

[12] Sites with frequent nocturnal temperature depressions due to CAP were identified for each study area using the empirical orthogonal function (EOF) techniques developed by *Lundquist and Cayan* [2007]. For each data set, we analyzed the daily minimum temperature, calculated as the lowest temperature recorded within a 24-h period starting at midnight. We defined daily minimum temperature at a point to be a function of (1) the mean annual minimum temperature from each station, $\bar{T}(\mathbf{x})$, primarily an elevation effect, (2) temporal deviations in the mean temperature across measurements within the domain, $\bar{T}'(t)$, primarily a synoptic-weather effect, (3) local spatial deviations that change through time, $\tilde{T}(\mathbf{x}, t)$, and (4) local instrument error, ε . Thus

$$T(\mathbf{x}, t) = \bar{T}(\mathbf{x}) + \bar{T}'(t) + \tilde{T}(\mathbf{x}, t) + \varepsilon. \quad (2)$$

The first two terms are well represented by existing techniques, where the slope of $\bar{T}(\mathbf{x})$ versus station elevation typically corresponds to the regional average lapse rate, and $\bar{T}'(t)$ corresponds to fluctuations in temperatures across all

stations due to variations in large-scale weather patterns. Essentially, these are region-wide positive or negative temperature anomalies. We analyzed local temperature patterns by first removing each station's long-term mean, $\bar{T}(\mathbf{x})$, and then removing the daily minimum temperature anomalies averaged across all stations, $\bar{T}'(t)$. The third and fourth terms were then decomposed into their principal spatial patterns of variation and their evolution through time using empirical orthogonal functions (EOFs) [*Beckers and Rixen*, 2003; *Preisendorfer*, 1988]. The EOFs are linear and orthogonal, such that a sum of each spatial component multiplied by its corresponding temporal score recreates the original temperature data set, and are normalized such that the variances of the spatial components sum to one, and the variances of the temporal components sum to the total variance of the original temperature record, $\tilde{T}(\mathbf{x}, t) + \varepsilon$. Within each of the three data sets examined, the dominant spatial mode of daily minimum temperature variations had a temporal component highly correlated with clear weather and weak winds, and a spatial component identifying locations with very low minimum temperatures during these events. Thus, the first EOF corresponded to CAP and accounted for 75%/59%/30% of the variance of $\tilde{T}(\mathbf{x}, t) + \varepsilon$ in the Rockies, Pyrenees, and Sierra Nevada, respectively. The percentage of the variance explained decreased as the area examined increased, because other factors such as variable exposure to air mass advection and slope orientation became increasingly important at larger domain sizes. The present analysis focuses only on CAP, and other modes of variation are not discussed.

[13] To summarize, EOFs decompose temperature variability into spatial and temporal weights, which become space-time series of temperature variation when multiplied together. For the first EOF, Figure 2 illustrates the spatial weights (Figure 2a), the temporal variations, also called principal components (Figure 2b), and two representative minimum temperature time series for the Pyrenees (Figure 2c). The two flat-bottomed valleys, Cerdagne and Capcir, exhibited strong CAP while the steeper-sided Conflent

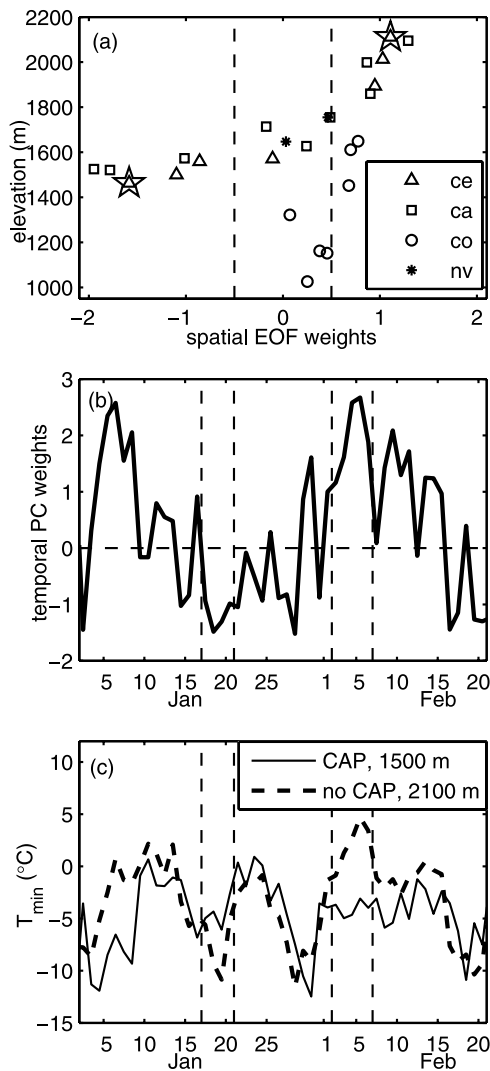


Figure 2. (a) First empirical orthogonal function (EOF), with negative weights corresponding to cold-air pooling (CAP) for the Cerdagne (“ce”), Capcir (“ca”), and Conflent (“co”) valleys. Sites not in any valley are marked “nv.” Vertical dashed lines at weights of -0.5 and 0.5 identify cutoffs classifying sites as CAP (<-0.5), no signal (-0.5 to 0.5), and no CAP (>0.5). (b) Two-month segment of the principal component (PC) time series corresponding to the spatial weights. (c) Original daily minimum temperature records from the two Cerdagne Valley sites marked with stars in Figure 2a for the same time period as in Figure 2b. The solid line has a negative weight and is prone to CAP, while the dashed line has a positive weight and does not experience CAP. Vertical dashed lines in Figures 2b and 2c identify example time periods with no CAP (17–21 January) and with strong CAP (2–7 February).

valley did not (Figure 2a). High principal component (PC) values (Figure 2b) indicate time periods when cold-air pools (CAP) were prevalent, i.e., sites with negative spatial weights had minimum temperatures 2° – 6°C colder than the regional average. Physically, sites with strong negative weights had greater temperature depressions than the regional average during a CAP event, while sites with

strong positive weights had warmer temperature anomalies than the regional average on those same days (an example of each is shown in Figure 2c). Sites with near-zero weighting had temperature anomalies close to the regional average during CAP events or had unsystematic anomalies. This could occur because these sites experienced slight cold-air pooling or infrequent cold-air pooling, whereas sites with strong positive weights very seldom experienced CAP temperature depressions.

[14] On average in the Pyrenees, some cold air pooling occurred, such that the site at 1500 m was colder than the site at 2100 m at times when the PC weight was near 0, i.e., when $\bar{T}(\mathbf{x}, t)$ was near 0 and $T(\mathbf{x}, t) \approx \bar{T}(\mathbf{x}) + \bar{T}'(t)$ (Figures 2b and 2c). Times with negative PC weights, such as 17–21 January 2004, identify periods with strong gradient winds and no CAP, when higher elevations were cooler than the valley bottoms. Positive PC weights indicate times with stronger than normal CAP, when the temperature inversion between CAP-prone locations and higher elevations was much stronger than usual, with a greater temperature depression in the valleys.

[15] Within each of the three data sets, high values of the principal components (PCs) of CAP patterns (Figure 3) were correlated with large-scale patterns of high pressure, clear skies, and weak gradient winds, as found by many other studies [Barr and Orgill, 1989; Clements et al., 1989; Gudiksen et al., 1992; Lundquist and Cayan, 2007]. The

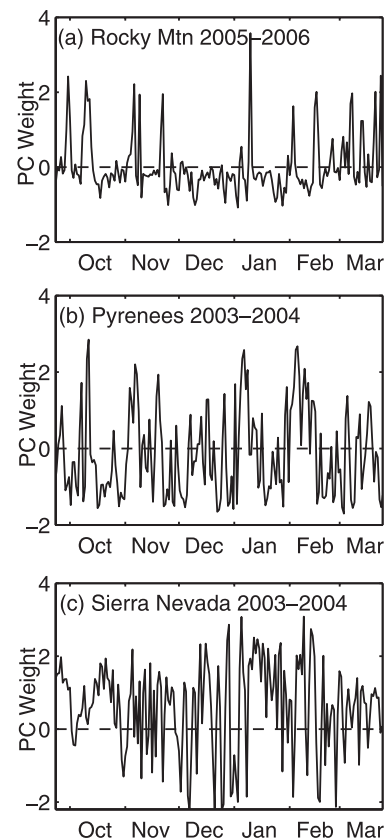


Figure 3. Time series of temporal weights (PCs) for (a) the Rocky Mountain data set, (b) the Pyrenees data set, and (c) the Yosemite, Sierra Nevada, data set during one winter.

Rocky Mountain site (Figure 3a) is on the eastern side of the Continental divide and because of its high mean elevation, is dominated by strong westerly winds, which generally prevent the formation of thermally forced air circulations. Thus, mean conditions describe a steep lapse rate and no CAP. The PC for this region hovers near zero except for distinct time periods of weak westerlies, when local circulations set up and result in significant cold-air pooling in flat valley bottoms. In both the Pyrenees (Figure 3b) and the Sierra Nevada (Figure 3c), such strong winds are not normal, as evidenced by PCs that oscillate between positive and negative values. Thus, moderate CAP occurs most nights, with time periods of no CAP or particularly strong CAP modulated by larger-scale circulation patterns.

[16] The standard deviation of the temporal variation in CAP was 0.75, 1.06, and 1.16°C for the Rocky Mountains, Pyrenees, and Sierra Nevada, respectively. Thus, for a site to experience $\pm 0.5^\circ\text{C}$ temperature oscillation when the temporal PC varies by one standard deviation, it would need a spatial weight magnitude of 0.67, 0.47, or 0.43, respectively, for the three study areas. For simplicity and generality, we used the average of these, 0.5, as a cutoff value for classifying CAP (Figure 2a) and then tested the sensitivity of the cutoff value for each study area (see section 4.4). We classified sites as “CAP” (EOF weight < -0.5), “no CAP” (EOF weight > 0.5), and “no signal” (EOF weights between -0.5 and 0.5). “No CAP” means sites show warmer temperatures than the regional average during CAP events, whereas “no signal” means that the EOF representing CAP has little or no influence on the temperature variance at these sites.

[17] To test the robustness of the EOF technique for identifying CAP sites, we ran the analysis for subsets of the sites with different time periods and groups of included sites. While the precise EOF weights changed slightly, the general classifications defined above (CAP, no CAP, and no signal) were consistent for most sites, with the exception of a few sites with weights very close to the cutoff value (see section 4.4).

3.2. Mapping Regions of CAP on a Digital Elevation Model (DEM)

[18] As discussed in the introduction, CAP occurs in concave or flat locations which have low elevations relative to the surrounding topography. First, we analyze topographic characteristics for each grid cell to identify flat or concave locations surrounded by higher land, which highlight potential CAP areas. These are necessary but not sufficient requirements for CAP, as a valley may drain, rather than pool, due to a downstream decrease in pressure. To account for draining valleys, we then calculate along-valley changes in the TAF and update the CAP classification along the valley bottoms.

3.2.1. DEM Mapping

[19] Many studies have developed algorithms for identifying landscape characteristics, such as flat valley bottoms and regions of likely sediment deposition, based on digital elevation models (DEMs) [e.g., *Gallant and Dowling*, 2003]. Following this prior work, we used DEMs with resolutions from 10 m to 100 m (Figures 4a and 5a for the Rocky Mountains and Pyrenees, respectively) to map out regions of low slope within depressions (valleys) in the landscape. Suitability was classified using three parameters.

[20] First, a standard algorithm (here, Matlab’s `gradientm.m`) was used to derive surface maps of slope (Figures 4b and 5b).

[21] Second, the relevant radius for a pixel’s position on the landscape was determined as half the average distance between two peaks separated by a valley within the region of interest, as in *Gallant and Dowling* [2003]. Then, the rank elevation of each pixel relative to the elevation of surrounding pixels within the square with the specified radius in each cardinal direction from the pixel was calculated to identify local ridges or valleys (Figures 4c and 5c). This calculated the percentage of surrounding grid cells lower than the given cell as a fraction of the total number of surrounding grid cells.

[22] Third, the curvature within the user-defined radius was calculated for each grid cell, according to the formula in *Liston and Elder’s* [2006] snow model:

$$cv = \frac{1}{4} \left(\frac{1}{2r} \left(z - \frac{(z_w + z_e)}{2} + z - \frac{(z_n + z_s)}{2} \right) + \frac{1}{2\sqrt{2}r} \left(z - \frac{(z_{sw} + z_{ne})}{2} + z - \frac{(z_{nw} + z_{se})}{2} \right) \right), \quad (3)$$

where cv is curvature at a pixel, z is the elevation of that pixel, r is the user-defined radius, $z_{w/e/n/s}$ is the elevation of a pixel a distance r to the west/east/north/south of the pixel, and $z_{sw/ne/nw/se}$ is the elevation of a pixel a distance r from the pixel in both directions specified, i.e., the four corners of a square (Figures 4d and 5d). The curvature used here is not just the second derivative of slope (which depends on the elevations of immediately adjacent pixels) but depends crucially on the user-specified radius to determine if a pixel is in a valley or on a ridge within the topographic landscape.

3.2.2. Topographic Amplification Factor: Draining Versus Pooling Valley Bottoms

[23] Once potential CAP sites are mapped, along-valley trends in TAF are calculated to determine which valleys are likely to pool and which are likely to drain. This is accomplished by defining a series of ordered latitude and longitude points that represent the axis of the valley, e.g., a stream vector. Using the direction from one point on the axis to the next lower point to define the down-valley direction (D), the algorithm calculates a valley profile perpendicular to D , between end points that are the distance of the user-defined radius in both directions (using Matlab’s `mapprofile.m`). The valley bottom is calculated as the elevation of the lowest point within the profile, which is often the original grid point if a stream vector was used. The valley top is defined as the lower of the two maximum elevations on either side of the valley bottom. The valley depth is the difference between these two elevations. The valley top width is defined as the distance between the points on each wall at the valley top elevation. The valley’s cross-sectional area is calculated for the valley area enclosed beneath the valley top using the trapezoidal rule. The topographic amplification factor (TAF), as outlined in equation (1), is calculated for each point along the valley’s axis, and the general trend in TAF along the valley is determined by eye. Where TAF is clearly decreasing down-valley, valley bottom values that were classified as CAP by the automated algorithm are reclassified as no signal.

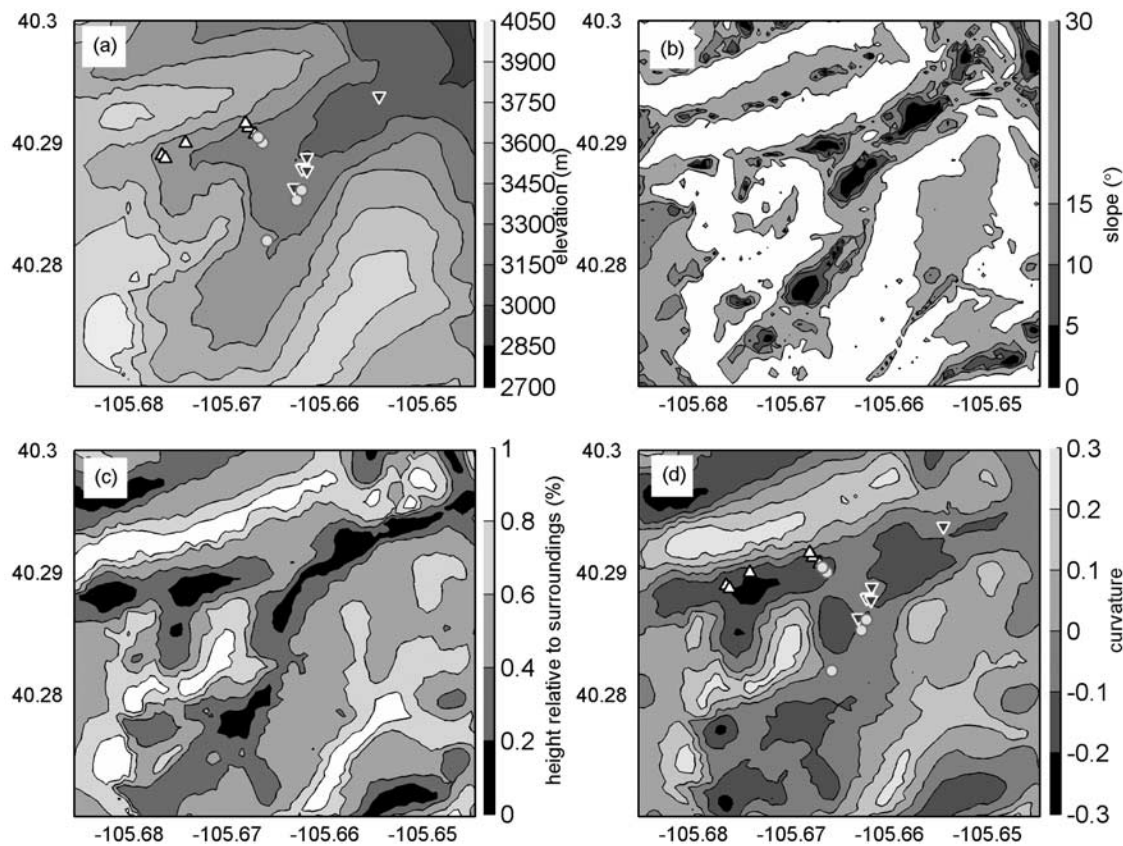


Figure 4. (a) Elevation, (b) slope, (c) percentile of elevation relative to surrounding elevations, and (d) curvature for the Loch Vale watershed in Rocky Mountain National Park, Colorado. In Figures 4a and 4d, inverted triangles indicate monitored locations that are sensitive to cold-air pooling, and triangles indicate areas that do not experience cold-air pooling. Circles represent sites without a clear signal for or against cold-air pooling.

Because of the additional computational power and subjective analysis required to assess the effects of along-valley changes in the TAF, statistics (detailed later) are calculated both with and without this correction.

3.3. Using CAP Maps for Temperature Interpolation

[24] Many techniques exist for interpolating temperature measurements across a landscape [e.g., *Dodson and Marks, 1997; Thornton et al., 1997; Stahl et al., 2006*]. These include: inverse distance weighting [*Dodson and Marks, 1997*], truncated Gaussian filters [*Thornton et al., 1997*], kriging [*Garen and Marks, 2005*], and multiple regression models [*Jarvis and Stuart, 2001*]. Various studies have concluded that the type of interpolation scheme or weighting strategy used is much less important than the location and representativeness of stations from which to interpolate [*Jarvis and Stuart, 2001; Stahl et al., 2006*].

[25] To test how important it is to identify regions of CAP for accurate temperature interpolation, we used a technique similar in principle to that developed by *Daly et al. [2002, 2007]*. In order to incorporate inversions and cold-air pools in the Parameter Elevation Regressions on Independent Slopes Model, *Daly et al. [2007]* identified local rises and depressions in the landscape within a 15 km radius and then weighted stations when interpolating such that stations with very similar topographic positions were weighted 100%,

and stations with quite different landscape positions were weighted by 0 and essentially ignored. Here, we imagine a situation where two temperature stations exist on the landscape, and temperatures at all other locations must be interpolated from these. For each study area, these two reference stations were forced to include one site identified by the EOF method as subject to CAP (EOF weight < -0.5) and one identified as not subject to CAP (EOF weight > 0.5).

[26] Before interpolation, temperatures at all of the sites were adjusted for elevation using a fixed lapse rate based on the average observed lapse rate, calculated by a regression of each site's average temperature versus its elevation, over the period of record. Average lapse rates were $9^{\circ}\text{C km}^{-1}$ for the Rocky Mountain study area, which has a dry, continental climate and an average lapse rate close to the dry adiabat, and $6.5^{\circ}\text{C km}^{-1}$ for the Sierra Nevada and the Pyrenees, which have moister, more maritime climates. To assess the importance of incorporating CAP in interpolation, temperatures were interpolated in four ways: (1) as the elevation-adjusted average of the two reference stations (linear interpolation); (2) using inverse distance weighting to place a higher weight on the proportionally closer station before averaging; (3) using the EOF-based classifications so that temperature was calculated for CAP sites based solely on the CAP reference station, for non-CAP sites based solely on the non-CAP reference station, and for no signal sites as

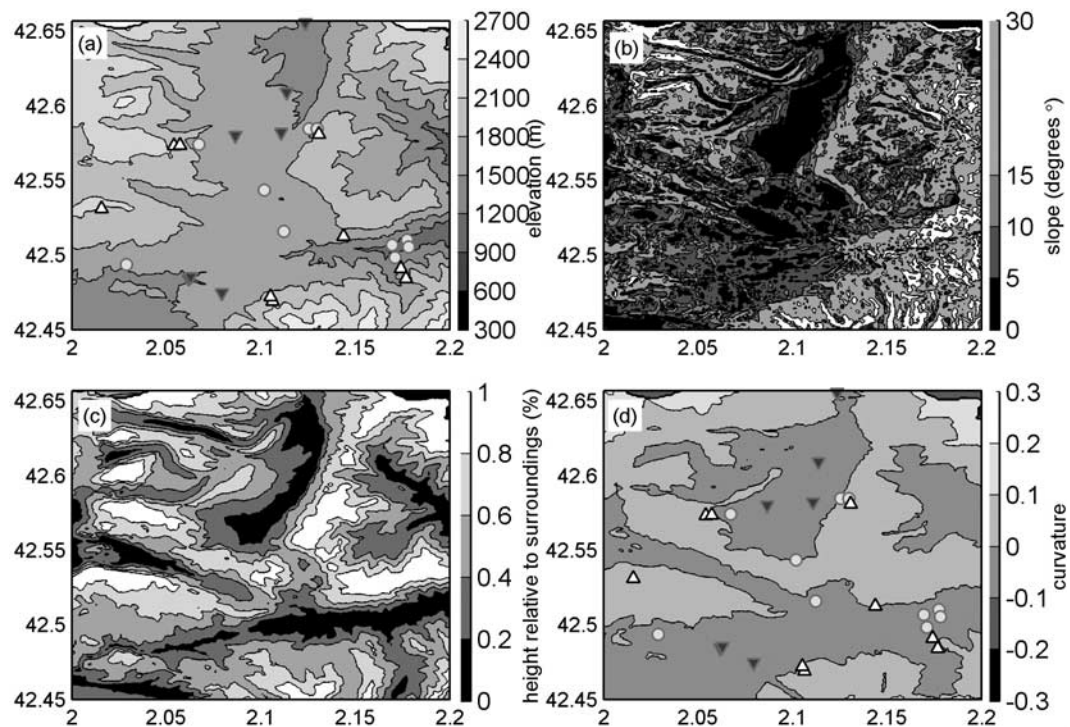


Figure 5. Same as Figure 4, but for the Eastern Pyrenees. Curved patterns near the top right and left corners of each graph are the edges of the watershed, where digital elevation model (DEM) information is not available.

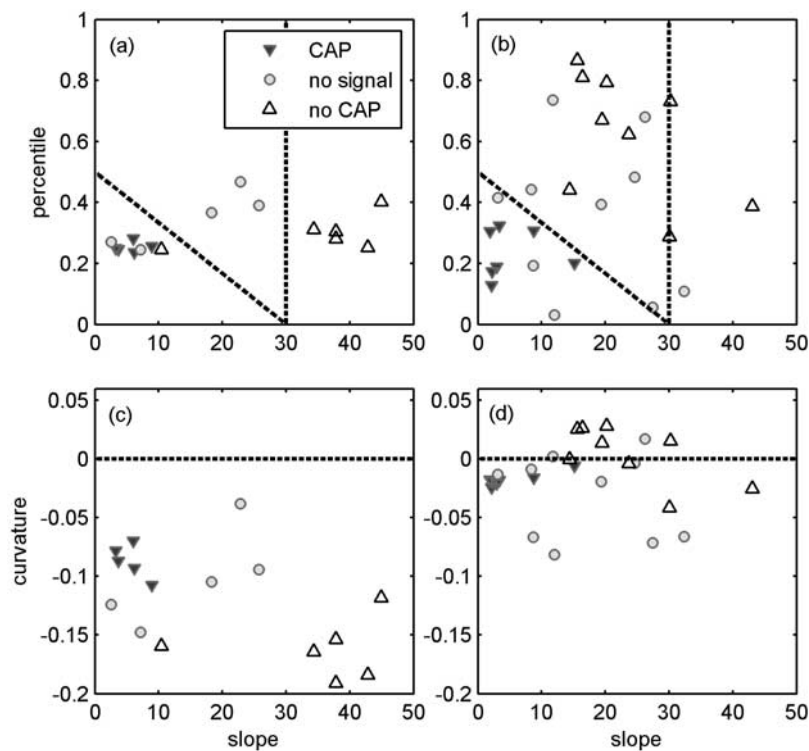


Figure 6. Scatterplots of (a, b) slope and percentile and (c, d) slope and curvature for stations classified as exhibiting CAP, no CAP, or no signal for Loch Vale, Rocky Mountain National Park (50 m resolution) (left), and the Pyrenees (100 m resolution) (right). Lines indicate the regions classified as CAP, no signal, and no CAP based solely on DEM characteristics.

Table 2. Interpolation Errors^a

RMSE (°C)	Linear Interpolation	Inverse Distance	EOF-CAP	Mapped CAP (Automated Only)	Mapped CAP (With Along-Valley TAF Correction)
Rocky Mountains (50 m, only 191 days with best data)	1.2	1.0	0.6	0.7	0.6
Pyrenees (100 m)	2.5	2.1	1.5	1.7	1.6
Sierra Nevada (100 m)					
Fixed radius	3.6	3.4	2.5	3.2	2.8
Variable radius	3.6	3.4	2.5	3.1	2.8

^aAbbreviations are as follows: CAP, cold-air pooling; EOF, empirical orthogonal function; RMSE, root-mean-square error; TAF, topographic amplification factor.

the average of the two reference stations, and thus no signal sites were treated the same as in method 1; (4) as in method 3, but using classifications based on CAP mapping (and not EOFs) using the mapping rules described in section 4.1.

[27] We cycled through all possible combinations of the two reference stations and calculated the resulting errors between interpolated and measured temperatures at all other sites for each pair of reference stations. RMSEs were calculated for each predicted site for each possible combination of the two reference stations for each of the four interpolation methods. RMSEs at each site were then averaged over the possible sets of reference stations to give an average error for each site.

4. Results: Developing and Testing the Automated Cold-Air Pool Mapping Algorithm

4.1. Rules for DEM-Based CAP Mapping, Colorado Rockies and Pyrenees

[28] We developed DEM-based rules for identifying CAP using the Rocky Mountain (50-m resolution DEM) and Pyrenees (100-m resolution DEM) data sets, which cover relatively small areas, and then tested the method on the Sierra Nevada data set, which included the greatest area and range of elevations. For each of the study areas, the radius (as defined in section 3.2) was determined as half the typical peak to peak distance, estimated by eye from the DEM. This was 375 m for the Rocky Mountain site, 3500 m for the Pyrenees, and about 1500 m for the Sierra Nevada.

[29] On the basis of scatterplots of CAP classification versus terrain parameters in the Rockies and Pyrenees (Figure 6), we set the following rules for initially designating sites as prone to CAP, never exhibiting CAP (no CAP), and ambivalent (no signal). Any location with slope greater than 30° and/or curvature greater than 0 was classified as never experiencing CAP (no CAP). Where the slope was less than 30°, the distinction between CAP and no signal was defined by the following line:

$$\text{Percentile} = -\frac{1}{60} \text{Slope} + \frac{1}{2}.$$

Locations with percentile and slope falling above this line were classified as having no signal, and locations with values below this line were classified as prone to CAP (see Figures 6a and 6b).

[30] As compared to the EOF-based classification for CAP, which relies on observed meteorological data for each location, these landscape rules alone correctly identified 14 out of 17 sites (82%) in the Rocky Mountains and 18 out of 26 sites (69%) in the Eastern Pyrenees. All

CAP locations were properly classified, but some no signal and no-CAP locations were improperly identified as prone to CAP.

[31] Figure 7 illustrates the resultant original maps of CAP classifications, how TAF varies along the valley axis,

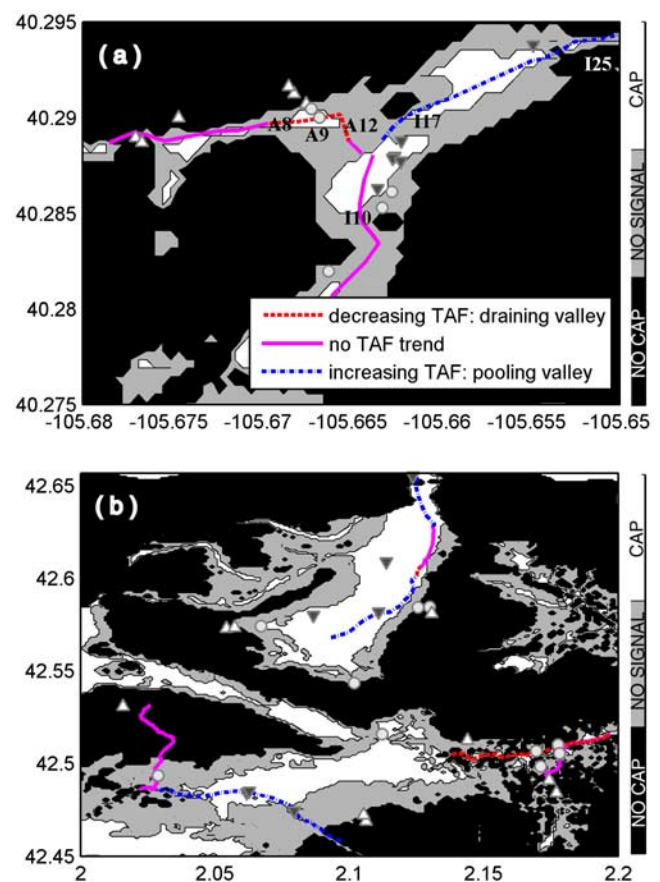


Figure 7. Selected areas of likely cold-air pooling (white), no cold-air pooling (black), and no clear signal (gray) across the landscape, based on the CAP mapping algorithm as described in section 4.1, for (a) the Loch Vale watershed in Rocky Mountain National Park, Colorado, and (b) the Pyrenees, France. Inverted triangles indicate monitored locations that are sensitive to cold-air pooling, and triangles indicate areas that do not experience cold-air pooling. Circles represent sites without a clear signal for or against cold-air pooling, as in Figures 4 and 5. Colored lines are streams along the major valley axes. A8, A9, A12, I10, I17, and I25 represent the locations of the cross sections shown in Figure 8.

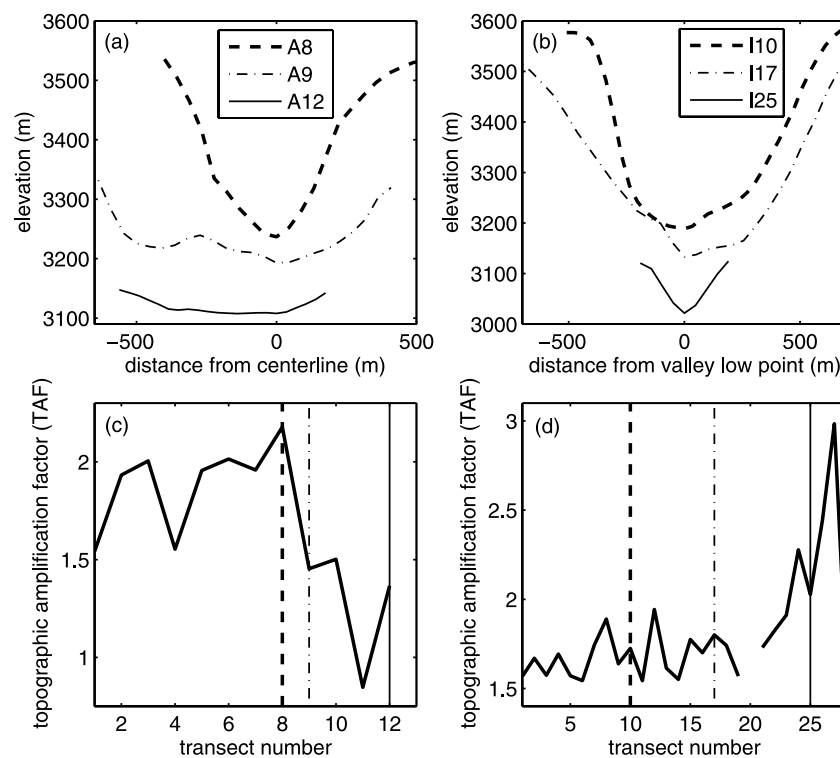


Figure 8. Valley cross sections and TAFs for (a, c) Andrew's Creek and (b, d) Icy Brook in Loch Vale, Rocky Mountains, illustrating that Andrew's Creek is subject to nocturnal winds that drain cold air, while Icy Brook, above the loch outlet, is subject to CAP. A8, A9, A12, I10, I17, and I25 in Figures 8a and 8b correspond to cross sections at locations marked in Figure 7, and to the transect numbers in Figures 8c and 8d.

and how these correspond with the EOF-based classifications at sensor locations. The TAF calculations, illustrated for Loch Vale in Figure 8, provide guidance for correcting the classification maps in Figure 7. For example, the TAF decreases along the lower portion of Andrew's Creek where the tributary valley widens to join the Icy Brook drainage (Figures 8a and 8c), indicating that more nocturnal cooling occurs in the upper reaches and that cold air will drain down-valley. On the other hand, TAF increases along the lower reaches of the Icy Brook drainage (Figures 8b and 8d) as the valley constricts near the outlet to the Loch. This results in more rapid localized nocturnal cooling down-valley and cold-air pooling.

[32] Locations labeled as CAP in valley bottoms that were identified as draining by the TAF were adjusted to a no signal classification. Specifically, 3 sites in Andrew's meadow along the Andrew's Creek drainage in Loch Vale and 2 valley stations along the Confluent drainage in the Pyrenees were reclassified as no signal. These valley adjustments increased the classification success rate to 16 of 17 sites (94%) in the Rocky Mountains and 20 of 26 sites (77%) in the Pyrenees, suggesting that along-valley factors help in mapping CAP locations.

4.2. Test Application: Yosemite National Park, Sierra Nevada, California

[33] The Sierra Nevada study area covered a much larger area than the data sets where the CAP mapping rules were developed, with varying peak-to-peak distances across

valleys and considerable areas where peak-to-peak distances could not easily be determined. For the initial test, an average radius of 1500 m (peak-to-peak distance of 3000 m) was selected for the entire terrain, although actual local radii, based on half the local peak-to-peak distances, varied from 300 m in high-altitude first-order streams to 3250 m near Tuolumne Meadows. (More discussion of radius is included in section 4.4.3.) A direct application of the CAP mapping rules outlined above (with $r = 1500$ m) correctly classified 30 out of 51 stations (59%) (Figure 9), assuming the EOF method represents the "truth." Adjusting for along-valley changes in TAF, such that draining valleys are classified as no signal instead of CAP, correctly classified 39 of 51 stations (76%), including one along Parker Pass Creek and two at the base of Lee Vining Canyon (Figure 10). Potential explanations for the remaining misclassifications are discussed in section 5.

4.3. Root-Mean-Square Error From CAP Mapping and CAP-Based Interpolation

[34] While maps of areas of likely CAP (as in Figures 7 and 10) help distinguish patterns on a landscape, the ultimate goal of CAP classification is to improve predictions of spatial temperature patterns. Therefore, to test how knowledge of CAP patterns (including any potential inaccuracies in these assumed patterns) influences temperature predictions, RMSE for predicted temperatures as compared to recorded temperatures were calculated for each study

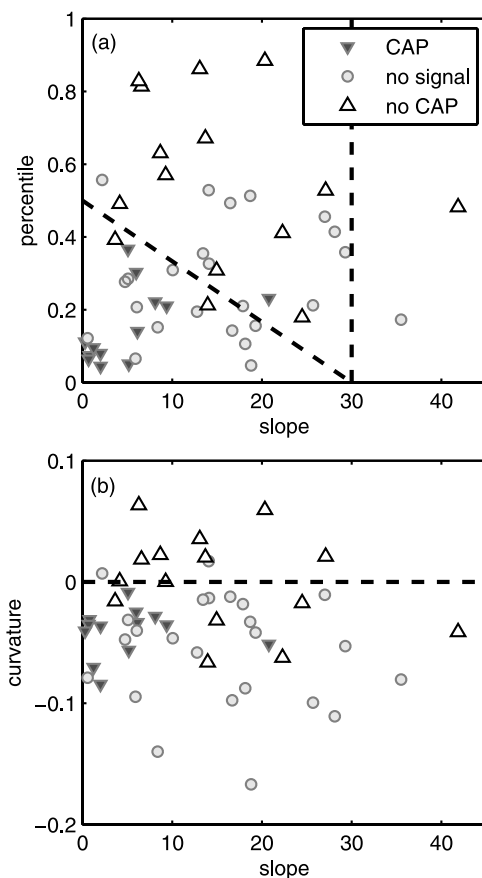


Figure 9. Same as Figure 6, but for 53 sites in the Yosemite National Park region, Sierra Nevada, California, for a fixed 1500-m radius.

area for each of the four interpolation methods listed in section 3.3.

[35] Averaged over all stations, RMSEs for simple linear interpolation increased as the size of the study areas increased, from 1.2°C in the Rocky Mountains, to 2.5°C in the Eastern Pyrenees, to 3.6°C in the Sierra Nevada (Table 2). This occurred because the number of reference stations was fixed at two, regardless of area. Inverse distance weighting (method 2) decreased the overall error, as compared to just elevation-adjusted linear interpolation (method 1), by 0.2°C in the Rocky Mountains and Sierra Nevada and by 0.4°C in the Pyrenees. Improvements at individual stations (not shown) ranged from 0 to 0.5°C. Averaged over all sites, EOF-based CAP classifications (method 3) reduced RMSEs (as compared to linear interpolation) by 0.6°C, 1.0°C, and 1.1°C in the Rocky Mountains, Pyrenees, and Sierra Nevada, respectively (Table 2). At individual stations, properly identifying CAP sites resulted in local RMSE decreases at CAP sites of 2–3°C over the 3-year analysis period in the Sierra, by 1–2°C over the 3-year period in the Pyrenees, and by 0.7–0.8°C over the 9-month period in the Rocky Mountains. Even with some misclassified stations, CAP mapping (method 4) outperformed simple linear interpolation on average in all three of the study locations, by 0.5°C in the Rocky Mountains and Sierra Nevada and by 0.8°C in the Pyrenees. Using mapped CAP

with the valley TAF correction reclassified the sites with the greatest temperature errors in the automated algorithm, resulting in error statistics close to those of the EOF-based classifications (Table 2). Figure 11 illustrates how RMSE differences between linear interpolated temperature and map-based CAP interpolated temperatures varied across the landscape for the three study areas for both the automated and TAF-corrected mapping algorithms. Most sites with increased errors in the basic algorithm had reduced errors once the valley was identified as draining rather than pooling by the TAF correction. Although most sites show significant error decreases, one site in a tributary to the Cerdagne (Figures 7b and 11d) and several sites near Tioga Pass (Figures 10 and 11f) still had misclassifications leading to increased errors. These are discussed in section 5.

4.4. Effects of Resolution, Radius Selection, and EOF Cutoff for Determining Curvature, Percentile, and CAP Classification

[36] For all three study areas, we tested the effects of DEM resolution, radius selection, and EOF cutoff on the calculated curvature, percentile, and resulting CAP classification.

4.4.1. DEM Resolution

[37] DEM resolution was relatively unimportant in site classification so long as the valleys and ridges of the area of interest could be accurately defined. For the Rocky Mountain study area, DEMs of 10, 20, 50, and 100 m resolution performed well, but a 500-m resolution DEM could not distinguish the relevant terrain. In the Pyrenees, on the other hand, a 500-m DEM was sufficient in the wider Cerdagne and Capcir valleys but resulted in more errors in the narrower Conflent Valley, resulting in 2 more misclassifications when a 3500 m radius was used.

4.4.2. EOF Cutoff

[38] Four sites in the Pyrenees had EOF weights very close to 0.5 (0.48, 0.46, 0.46, and 0.38) that were originally classified as no signal. Of these, the map-based algorithm classified one (with a weight of 0.46) as no signal and three as no CAP. Changing the EOF classification cutoff to 0.35 reclassified these four sites as no CAP, resulting in three more matching EOF- and map-based classifications, thus correctly matching 21 of 26 (81%) of the sites with the automated algorithm and 23 of 26 (88%) with the TAF correction. Ten sites in the Sierra study area had EOF weights with absolute values between 0.5 and 0.35. However, changing the cutoff to 0.35 only reconciled the classification at about half the sites, with negligible net changes to the statistics. The Rocky Mountain data set did not have any sites with EOF absolute values between 0.35 and 0.5 and thus was less sensitive to the EOF cutoff. In most cases of sites with EOF values near the cutoff, RMSEs did not change much when the site's classification was switched.

4.4.3. Radius

[39] In all circumstances, the greatest sensitivity in the success of CAP mapping was to the selected radius, where the radius equal to half the mean peak to peak distance provided the best results. Doubling or halving the radius led to two to three more misclassified sites in each study area. For regions with clear valleys, such as the Loch Vale Watershed in the Rocky Mountains, the Eastern Pyrenees, and much of Yosemite in the Sierra Nevada, the mean peak to peak distance across the valley can easily be estimated

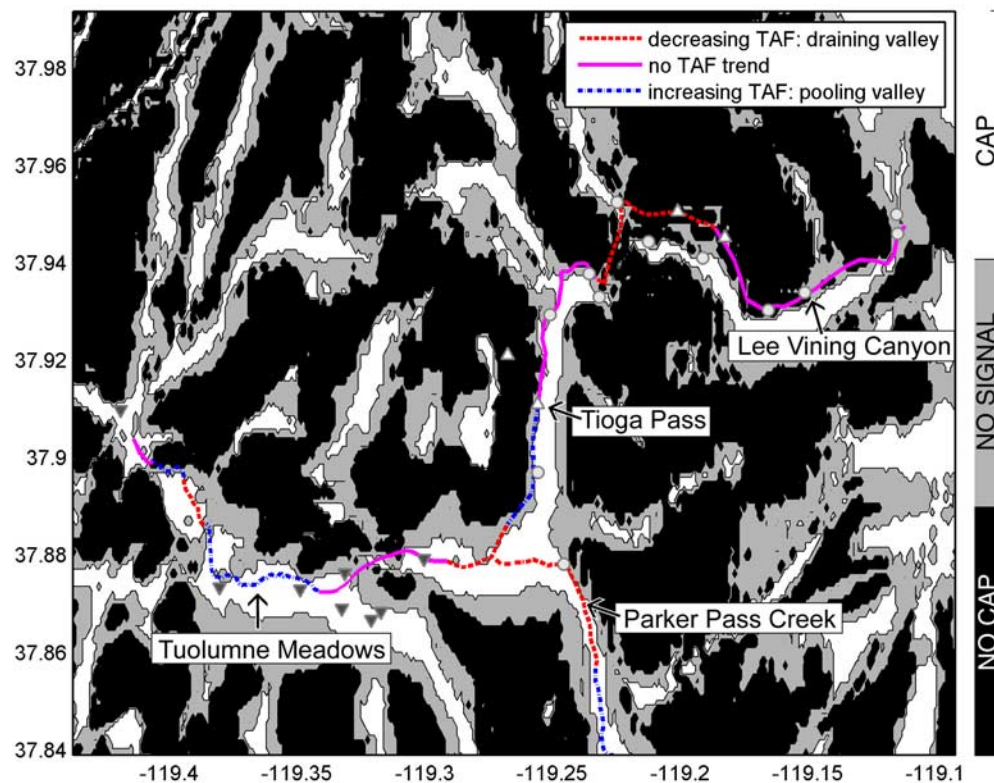


Figure 10. Same as Figure 7, but for a subset of the Sierra Nevada study area.

from a map or DEM. For larger areas with a dendritic drainage pattern, as in the Sierra Nevada, the peak to peak distance is typically smaller for higher elevation, headwater streams and is larger for lower, higher-order streams. Thus, the universal application of one radius may be inappropriate.

[40] We reran the model for the Sierra Nevada area using the DEM to define an appropriate radius separately for each sensor location. The appropriate radius was hard to determine for many stations along the Tioga Road, which passes through rolling terrain on the western slope and down a steep escarpment on the eastern side (Figure 1a). Using a locally determined radius corrected the classification at three sensor locations in the Upper Merced drainage that fell within a narrow canyon with a small peak-to-peak distance (<800 m). However, this approach led to new misclassification at two stations on the eastern slope where an appropriate radius was not clear. Therefore, although locally defining the radius based on valley width can improve classification in river valleys, other criteria need to be developed for nonfluvial terrain.

5. Discussion: Reasons for Errors and Possible Improvements

[41] The mapping algorithm presented (even with the TAF modification) here is designed to be simple, widely applicable given any DEM, and easy to apply. However, several processes important to the intensity of CAP have been ignored, resulting in misclassifications. Problems with map-based CAP classification resulted from the following:

(1) EOF weights too close to the cutoff, as described for the Pyrenees and Sierra Nevada, (2) local slope differing from the DEM-calculated slope due to poor resolution of microtopography, and (3) saddle-shaped topography near a windy mountain pass, as shown for the Tioga Pass region of Yosemite (Figure 10). The first is an artifact of the set of sites being analyzed, since it refers to the CAP strength relative to other sites in the study area. Only the last two problems resulted in increased RMSEs at misclassified sites, and these are discussed below.

[42] Several sites in the Pyrenees and Sierra Nevada located on steep slopes just above meadows mapped as having a gentle slope by the DEM were improperly classified. Conversely, one sensor in Yosemite on a very flat location next to a near-vertical cliff was mapped with too steep a slope and thus misclassified. All of these sites were near the edges of DEM classification areas, and small errors in position resulted in an incorrect classification. This suggests that while overall mapping is robust with respect to microtopography, (for example, this would not be problematic for an area-integrated process such as regional snowmelt), classification of a specific temperature sensor should be made with care to its precise slope and position on the landscape.

[43] The largest errors in the Sierra Nevada study area were sites next to alpine lakes near a mountain pass (Figures 10 and 11). The lakes were flat locations in saddle-shaped topography that were low enough compared to surrounding locations to be classified as CAP. The TAF was relatively constant through this region. However, the mountain pass and escarpment just east of it channeled the

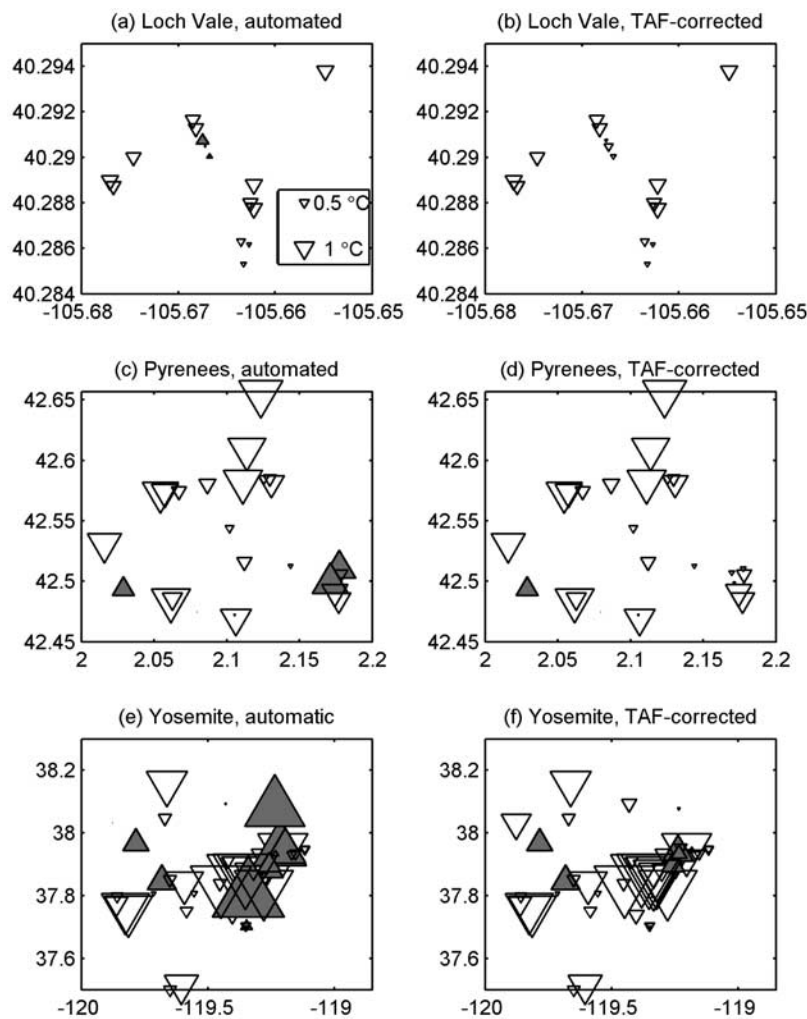


Figure 11. Mapped root-mean-square error changes for mapped CAP-based interpolation versus ordinary linear interpolation for (a, b) Loch Vale, Rocky Mountains, (c, d) Pyrenees, and (e, f) Yosemite, Sierra Nevada. Triangles show an increase in error. Inverted triangles show a decrease in error. Triangle size is proportional to the size of the error increase or reduction, as shown in Figure 11a, for all panels.

local wind, resulting in increased turbulence and mixing. This is likely what prevented CAP formation in these locations.

[44] Overall, most of the misclassifications are well understood, and DEMs can be used for spatial temperature predictions across many mountain landscapes.

6. Conclusions

[45] For distributed snowmelt or ecological modeling, or for depicting spatial patterns of rain versus snow across a landscape, accurate spatial descriptions of temperature are paramount. This article provides an algorithm for identifying which areas of a landscape are likely to be prone to cold-air pooling. The algorithm uses DEMs at resolutions of 10 m to 500 m to calculate slope, percentile (rank of elevation relative to surrounding terrain), and curvature at each pixel. The percentile and curvature depend on a specified radius, which should be estimated as half the average peak-to-peak distance measured from the DEM.

Cold-air pools (CAP) occur in local depressions, i.e., low percentile and negative (concave) curvature, with flat slopes. Additional consideration of downstream changes in the topographic amplification factor (TAF), although more computationally intensive, improves the mapping considerably by clarifying whether valley bottom locations are pooling or draining. Errors occurred where the micro-topographic character of a specific sensor was different from the slope/curvature of its pixel on the DEM, or where winds were channeled through a mountain pass, resulting in increased turbulence that prevented CAP formation. However, despite some errors in classification, using CAP maps to interpolate temperature across the landscape improved the overall temperatures predicted for each of the three study areas.

[46] The automated CAP mapping techniques presented here can identify where to put a weather station, or when to be careful regarding interpreting a weather station's records. Specifically, CAP maps could be used before deploying

instruments in mountainous regions, to allow stratified sampling. Deploying sensors so that one is in a CAP region and one is not and using the CAP-based interpolation method described here provides approximately 1°C improvement in distributing $T_{\min}$, on average, over simply averaging the temperature at any two stations. The quality of the data and the microtopography at the two stations should be checked very carefully, as low-quality data or a misclassification at either station will bias the predictions for that area.

[47] Owing to accessibility, existing temperature measurement locations are heavily weighted toward valley floor locations, which means that many are in cold air pools many nights. When these sites are used to extrapolate mean daily temperatures to higher elevations, they may misrepresent higher-elevation temperatures as being too cool. For snowmelt models and projections of climate sensitivity, this may underestimate melt. The frequency of CAP events depends on the frequency of calm, clear nights, which changes with synoptic weather patterns. Thus, a shift in regional circulation patterns could cause misrepresentative temperature trends to appear in records at CAP locations. If the only temperature available for a region is in a CAP location, care should be taken to adjust nighttime minima up at non-CAP locations on clear, calm nights. The maps presented here can identify where that adjustment should be made, but a separate method would then be needed to calculate the precise magnitude of the adjustment.

[48] Across the globe, nighttime temperatures have been warming much more rapidly than daytime temperatures [Karl et al., 1993]. Many biological organisms, such as bark beetles, crickets, piñon mice, and pine trees, are limited by nighttime mean temperatures or extreme minima [Beatley, 1975; Tenow and Nilssen, 1990; Virtanen et al., 1998; Wearne and Morgan, 2001], leading to inverted tree lines and colonies of more cold-tolerant species, such as pikas, being found in CAP-prone locations. Thus regions with frequent cold-air pools may prove to be biological refuges in a warmer world. The techniques described here provide a backbone for designing ecological studies to test this hypothesis.

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